

Multiple imputation method of missing credit risk assessment data based on generative adversarial networks

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ABSTRACT

Credit risk assessment is critical for loan approval and risk management of banks. However, the problem of missing credit risk data may greatly reduce the effectiveness of the assessment model. Therefore, constructing a data imputation method for accurate missing data prediction is quite beneficial. Typically, building an effective imputation model is very challenging due to the high missing rate and complex arbitrary missing pattern of datasets in credit risk assessment. In this paper, a novel imputation method named as Multiple Generative Adversarial Imputation Networks (MGAIN) is proposed. Specifically, we first randomly select multiple attribute subsets instead of the whole attributes such that more complete samples can be generated. Then, the missing data in each attribute are imputed by using generative adversarial imputation networks (GAIN) which fully considers the relationships among missing values by combining neural network and adversarial learning. The proposed subset selection and multiple imputation strategy not only simplify the network structure of GAIN but also reduce the demand for data. Finally, a weighted average method is presented to synthesize multiple results of each missing attribute value to further improve the accuracy. The experimental results on real-world data demonstrate that the proposed method is superior to other popular imputation methods.

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1. Introduction

Credit risk assessment plays a key role in developing correct policies and strategies for banks [1] or some micro enterprises related to microcredit programs [2], and sound credit decisions contribute to the sustainable development of financial markets [3]. However, the collected credit risk assessment data is often missing due to the omission of bank customers, the negligence of collection staff, data fraud to avoid risk, etc. This problem seriously affects the prediction accuracy of the follow-up assessment model [4]. For credit risk assessment, the missing rate of datasets is usually high and the missing pattern is relatively complex, which also makes the imputation of missing data challenging. Therefore, this work aims to investigate an effective data

imputation method with high accuracy to predict the missing credit risk assessment data.

Before formal study, we should know the mechanism of generating missing data, which can be divided into three categories: missing completely at random (MCAR), missing at random (MAR), and missing not at random (MNAR) [5]. MCAR means that the missingness is completely unrelated to any variable in the dataset [6]. In this case, any missing data handling method can be applied since there is no risk of affecting the data distribution [7]. MAR means that the missingness is related to some other observed variables [8]. At this time, direct deletion or partial processing methods may lead to biased results [9]. MNAR means that the probability of missing data depends on the value itself [10], which may lead to the loss of useful information. In this paper, we provide analysis under the assumption that missing mechanism of data is MCAR.

In the past few years, many promising methods have been proposed for imputing the missing credit risk assessment data, which can be grouped into three categories. The first category is the statistical methods. The simplest statistical imputation method is to replace missing data with attribute mean [11,12],

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mode [13], or the data with the highest frequency for interpolation [14]. These methods are relatively easy to implement, but they may also easily cause resource waste or data distribution distortion [15,16], which leads to bias in parameters, results and inferences of the subsequent evaluation model [17]. In addition, regression imputation method [18] is also commonly applied for the data imputation based on statistics, including linear regression [19], logistic regression [20] and so on. For example, Hamzah et al. [21] combined robust random regression imputation and multiple linear regression to impute the missing data in daily streamflow datasets, and achieved excellent results. However, regression imputation method has a great limitation, that is, when the complete dataset for training is very small, the performance of the trained regression model is very poor, resulting in the decline of imputation performance. Moreover, for linear regression or logistic regression, its restrictive assumption of linearity makes it unable to be used in real-world situations where the relationship between multivariates is rather complex [22].

The second category is the machine learning methods. With the development of artificial intelligence technology, machine learning has gained rising attentions for their satisfactory results. For example, the imputation method based on K-nearest neighbor (KNN) algorithm [23] estimates the missing values by combining the columns of K-nearest records chosen based on a given similarity measure [24]. It is suitable for both numerical and categorical data as long as an appropriate distance function is selected. However, the performance of this method is limited when the amount of missing data is large since it only uses complete cases to fill missing values [25]. In addition, the selection of appropriate similarity metric and an optimal value for k are also nontrivial [26]. Support vector regression (SVR) is optimized from support vector machine (SVM), which is also a classic imputation method [27]. Tensor-based imputation method is applied to various fields to solve practical problems, such as the estimation of missing velocity data in the traffic network [28,29]. Nikfalazar et al. [30] proposed a novel imputation method by integrating the merits of decision trees and fuzzy clustering into an iterative learning approach, and proved its prediction accuracy and reliability. Su et al. [31] proposed an imputation method based on statistics and machine learning, which combines global trend and local variation. The test results on public dataset show that the method outperforms the commonly used methods. Although the imputation methods based on machine learning have played a great role, once the amount of data is greatly increased or the relationship between attributes is more complex, the capability of these methods to reflect the mapping relationship between attributes may be limited.

The third category is the deep learning methods. The advent of the era of big data and the improvement of computer technology greatly promote the development of deep learning. Such methods are more and more used in the field of data imputation. For example, the imputation method based on neural network such as back propagation (BP) neural network [32,33], is more powerful due to its non-linear approximation ability [34]. However, when facing serious data missing, BP neural network is easy to overfit. Lin et al. [35] applied deep neural networks, namely multi-layer perceptron [36] and deep belief network [37], to impute the missing values. The experimental results show that these two models are significantly better than the baseline imputation methods based on mean, KNN and SVM. In 2018, Yoon et al. [38] first proposed generative adversarial imputation networks (GAIN), and achieved good results in filling missing data. GAIN combines both the generative and discriminative models in an adversarial manner to excel the imputation tasks [39]. The ultimate purpose of this method is to make the distribution of new samples generated by the generative model very close to the distribution of original

data, so as to achieve better imputation effect [40]. Yang et al. [41] proposed a spatio-temporal learnable bidirectional attention generative adversarial networks to impute the missing traffic data, and the model was well applied to the taxi GPS dataset. Awan et al. [42] proposed to impute missing data based on class-specific characteristics by adapting the popular conditional generative adversarial networks, and achieved superior performance. Deep learning model can better capture data hiding patterns and underlying dynamics, and is more suitable for dealing with complex datasets.

In general, most of the above methods have achieved excellent results in data imputation, but they still have these two deficiencies. (1) The relationships among missing values cannot be fully utilized, which greatly reduces the accuracy of imputation. (2) The missing data of multiple variables cannot be imputed at the same time, resulting in the decrease of imputation efficiency. Although some deep learning models have solved these two problems, such as GAIN model, this model also has the following two limitations. (1) The performance of GAIN is greatly affected by the dataset. When the number of attributes with missing data is large, in order to get more accurate imputation results, GAIN needs two conditions, i.e., a large amount of data and more complex network structure in terms of more network layers and the number of neurons. We can give a more vivid example. Assume that a dataset contains 1000 samples and 10 attributes. If only two attributes in the dataset contain missing values, and the missing rate of each attribute is 10%, the complete data accounts for 81.0% of the total data. However, if all 10 attributes in the dataset contain missing values, and the missing rate of each attribute is 10%, the complete data accounts for 34.9% of the total data. The great reduction of complete data will definitely affect the training of the model. In order to achieve satisfactory imputation results, it is necessary to increase training data and the capacity of the model, which are often difficult to be met in real-world practice. (2) The method of determining the final imputation result by single imputation is not properly considered. Single imputation often ignores the uncertainty about the true value of imputed variables [43], which may decrease the accuracy of imputation results.

To address the above two issues in GAIN, we propose an imputation method called Multiple Generative Adversarial Imputation Networks (MGAIN), for accurate prediction of the missing data in credit risk assessment. Fig. 1 illustrates the overview of the proposed MGAIN model, where $A_1, A_2, \dots, A_n$ represent n attributes, respectively. Specifically, MGAIN includes four steps: (a) Selecting multiple attribute subsets. We fully consider the relationships between each attribute and randomly select multiple subsets from all the attributes. (b) Filtering subsets. After training and testing on the complete dataset using GAIN, the subsets with poor prediction performance are filtered out. (c) Predicting the missing data. Each filtered subset uses GAIN to achieve prediction on the missing dataset. (d) Synthesizing multiple results. Multiple results of each missing attribute value are integrated by using weighted average method to obtain the final imputation results.

In summary, our proposed MGAIN has three major advantages. (1) Subset selection and multiple imputation based on GAN effectively simplifies the network structure of GAIN and increases the amount of available data for training. (2) A weighted fusion method of multiple imputation results is provided in this paper, which can further improve the accuracy of the final imputation value. (3) The application scope of generative adversarial networks (GAN) is expanded. To the best of our knowledge, it is the first time that GAN is applied to impute the missing data in credit risk assessment. Credit risk assessment data is often seriously missing and the mapping relationship between data is complex, which leads to the difficulty of imputation. GAIN can cope with this situation better, and it can fully consider the

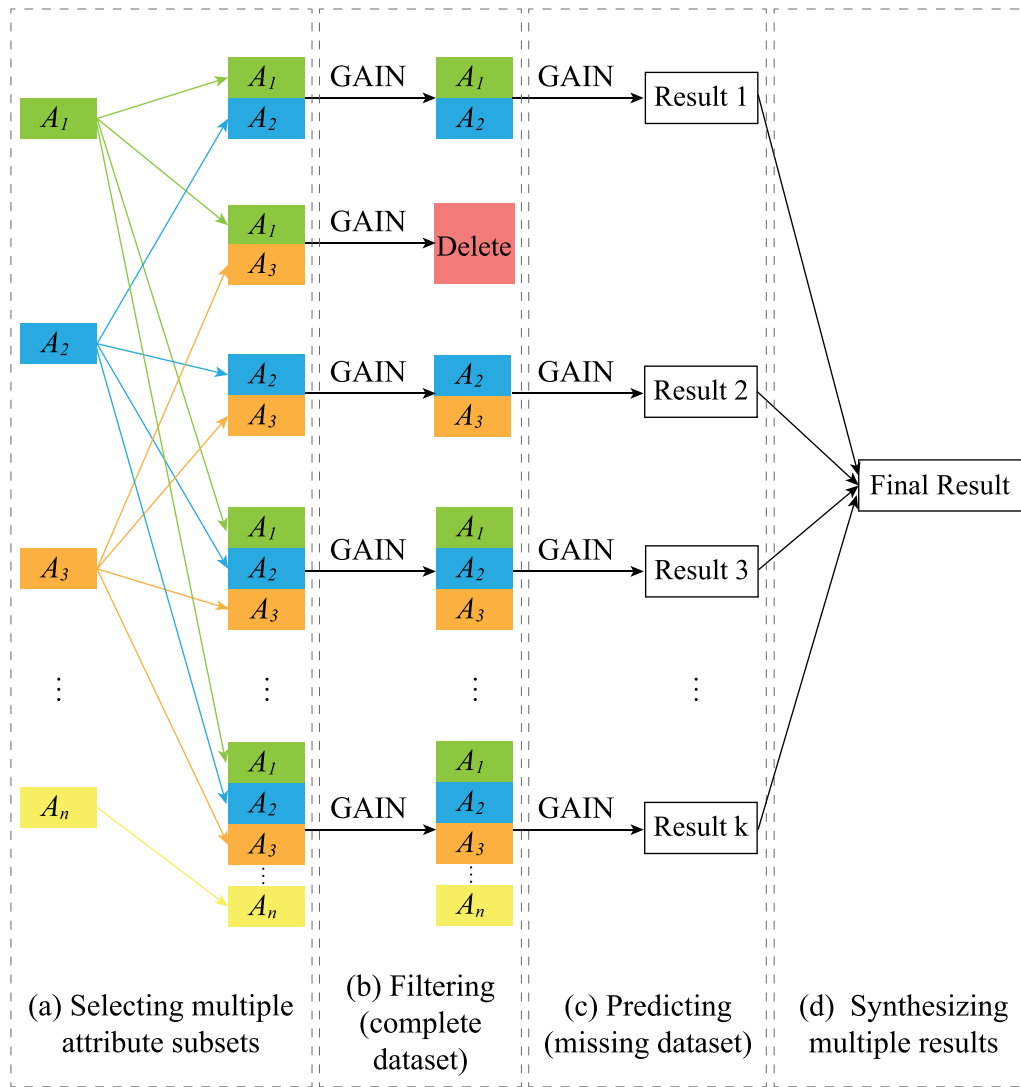


Fig. 1. Overview of MGAIN.

relationships among missing values to complete the imputation of multivariate missing model at the same time. In order to verify the effectiveness of MGAIN, experiments are designed by using the real data of bank credit risk assessment. The experimental results demonstrate that MGAIN has superior performance over traditional GAIN and other popular imputation methods.

We reemphasize the difference between our study and what has already been done in the literature. Different from other studies on imputation methods based on traditional models, such as linear regression, BP neural network, our study adopts a deep learning model GAIN, which can better model the relationships among missing values. Further, different from the imputation methods based on GAIN model, we propose subset selection and multiple imputation to simplify the network structure, reduce the demand for data, and greatly improve the accuracy and stability of prediction results.

The rest of paper is organized as follows. Section 2 briefly introduces GAN and GAIN. In Section 3, we provide a detailed description of MGAIN. The effectiveness of MGAIN is demonstrated by several experiments in Section 4. Finally, the conclusions of our study are given in Section 5.

2. Generative Adversarial Imputation Networks (GAIN)

GAN [44] is a hot research topic of deep learning in recent years, and has shown outstanding achievements in many fields [45–47]. Fig. 2 shows the structure of a typical GAN and Fig. 3 shows its application in data imputation, i.e., GAIN. Typically, a GAN framework consists of two separate networks, i.e., the generator and discriminator, as shown in Fig. 2. GAN was initially proposed to generate samples based on the given random noise. Then, the adversarial training of generator and discriminator is performed so that the generator learns how to generate the more realistic fake samples which can fool the discriminator [48], whereas the discriminator learns how to distinguish between the generated high-quality fake samples and real samples [49]. After a certain period of training, discriminator finally cannot distinguish the authenticity of the samples generated by the generator.

In view of the great success of GAN in image inpainting [50,51] and its similarity with data imputation, the research on GAN-based data imputation is emerging. For example, missing data imputation using GAN, i.e., GAIN [38], was first proposed in 2018. It can deal with the unique characteristics of the imputation problem, and has achieved good result.

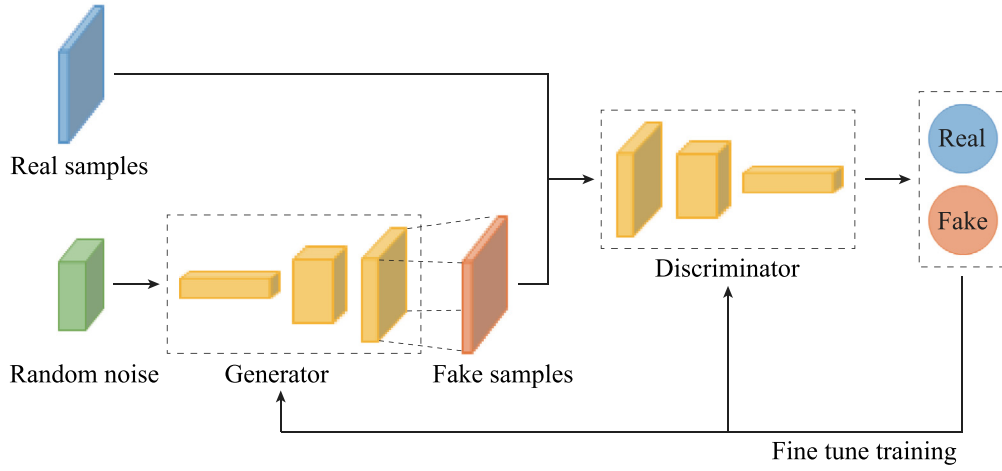


Fig. 2. Structure of a typical GAN.

Fig. 3 shows the structure of GAIN. The generator in GAIN learns the distribution $p_{data}(x)$ over data x , and then outputs the completed imputation results. The discriminator recognizes which components are actually observed and which are imputed. The minimax objective function is

$$\min_G \max_D V(D, G) = E_{x \sim p_{data}(x)} [\log D(x)] + E_{z \sim p_z(z)} [\log(1 - D(G(z)))]. \quad (1)$$

In general, the iterative method is used to solve the minimax optimization problem in (1). First of all, optimizing the discriminator with a fixed generator until the loss function converges. Then, optimizing the generator with the newly updated discriminator until the loss function converges. Among them, the loss function can be in the form of cross entropy, or be defined according to the specific task.

Compared with some classical imputation methods, GAIN shows its obvious advantages. GAIN can not only make full use of all the relationships among the missing values to make the imputation results more accurate, but also impute multiple missing variables at the same time, which greatly improves the imputation efficiency.

However, GAIN also has some room to be improved. First, when there are many attributes with missing data, GAIN relies on complex network structure and a large amount of training data. These conditions are difficult to be met in real-world practice. In addition, GAIN belongs to single imputation, which makes the final results unstable and imputation accuracy not enough. To address these issues, we propose an improved model which will be presented in Section 3.

3. Multiple Generative Adversarial Imputation Networks (MGAIN)

Aiming to deal with the limitations of GAIN, we propose MGAIN, which can greatly improve the imputation accuracy and efficiency. Fig. 4 shows the specific flow chart of MGAIN.

For the convenience, we assume that there are m samples ($S_1, S_2, \dots, S_m$) and n attributes ($A_1, A_2, \dots, A_n$) in a missing dataset, as shown in Fig. 4 (NAN indicates missing data).

First of all, multiple attribute subsets are randomly selected from all the attributes, and each subset contains at least two attributes. We can get the attribute subsets $\mathbf{A} = \{A^{(2)}, A^{(3)}, \dots, A^{(i)}\}$ ($2 \leq i \leq n, i \in \mathbf{Z}$), where $\mathbf{A}$ is the total set of subsets and $A^{(i)}$ is the subsets with i attributes, which can be specifically expressed

as

$$\begin{cases} A^{(2)} = \{\{A_1, A_2\}, \{A_1, A_3\}, \dots, \{A_{t_1}, A_{t_2}\}\} \\ (1 \leq t_1 < t_2 \leq n) \\ A^{(3)} = \{\{A_1, A_2, A_3\}, \{A_1, A_2, A_4\}, \dots, \{A_{t_1}, A_{t_2}, A_{t_3}\}\} \\ (1 \leq t_1 < t_2 < t_3 \leq n) \\ \dots\dots\dots \\ A^{(i)} = \{\{A_{t_1}, A_{t_2}, \dots, A_{t_i}\}\} \\ (1 \leq t_1 < t_2 < \dots < t_i \leq n, t_1, t_2, \dots, t_i \in \mathbf{Z}) \end{cases} \quad (2)$$

From (2), there are at most C_n^i subsets in the set $A^{(i)}$, and all subsets are at most $2^n - n - 1$ in total.

In order to elaborate the first step of our method more intuitively, we take the attributes $A_1 - A_4$ as an example. Through permutation and combination, 11 attribute subsets in set $\mathbf{A} = \{A^{(2)}, A^{(3)}, A^{(4)}\}$ are listed as follows:

$$\begin{cases} A^{(2)} = \{\{A_1, A_2\}, \{A_1, A_3\}, \{A_1, A_4\}, \\ \{A_2, A_3\}, \{A_2, A_4\}, \{A_3, A_4\}\} \\ A^{(3)} = \{\{A_1, A_2, A_3\}, \{A_1, A_2, A_4\}, \\ \{A_1, A_3, A_4\}, \{A_2, A_3, A_4\}\} \\ A^{(4)} = \{\{A_1, A_2, A_3, A_4\}\}. \end{cases} \quad (3)$$

Then, multiple attribute subsets need to be filtered out of the complete dataset to discard the subsets with poor prediction performance. The complete dataset is trained and tested by using GAIN, and then the prediction accuracy of each attribute subset can be obtained, which is noted as $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_l)$ ($1 \leq l \leq 2^n - n - 1, l \in \mathbf{Z}$). We select the subsets with high prediction accuracy according to α , and note them as $\tilde{\mathbf{A}} = \{\tilde{A}^{(2)}, \tilde{A}^{(3)}, \dots, \tilde{A}^{(i)}\}$ ($2 \leq i \leq n, i \in \mathbf{Z}$), where $\tilde{\mathbf{A}}$ represents all the subsets with better prediction effect, and $\tilde{A}^{(i)}$ represents the subsets with i attributes.

The third step of our method is to predict the missing values on the missing dataset by using GAIN. The filtered subsets with high prediction accuracy are directly used for prediction. Suppose there are k ($1 \leq k \leq l, k \in \mathbf{Z}$) attribute subsets left for prediction. Multiple imputation results for each missing value can be obtained, noted as $y^{(1)}, y^{(2)}, \dots, y^{(k)}$.

The specific structure of GAIN used in both filtering and predicting in MGAIN is shown in Fig. 3. We define the two loss functions in detail. The loss function 1 in Fig. 3 is defined as

$$L_D(m, \hat{m}) = - \sum_i [m_i \log(\hat{m}_i) + (1 - m_i) \log(1 - \hat{m}_i)], \quad (4)$$

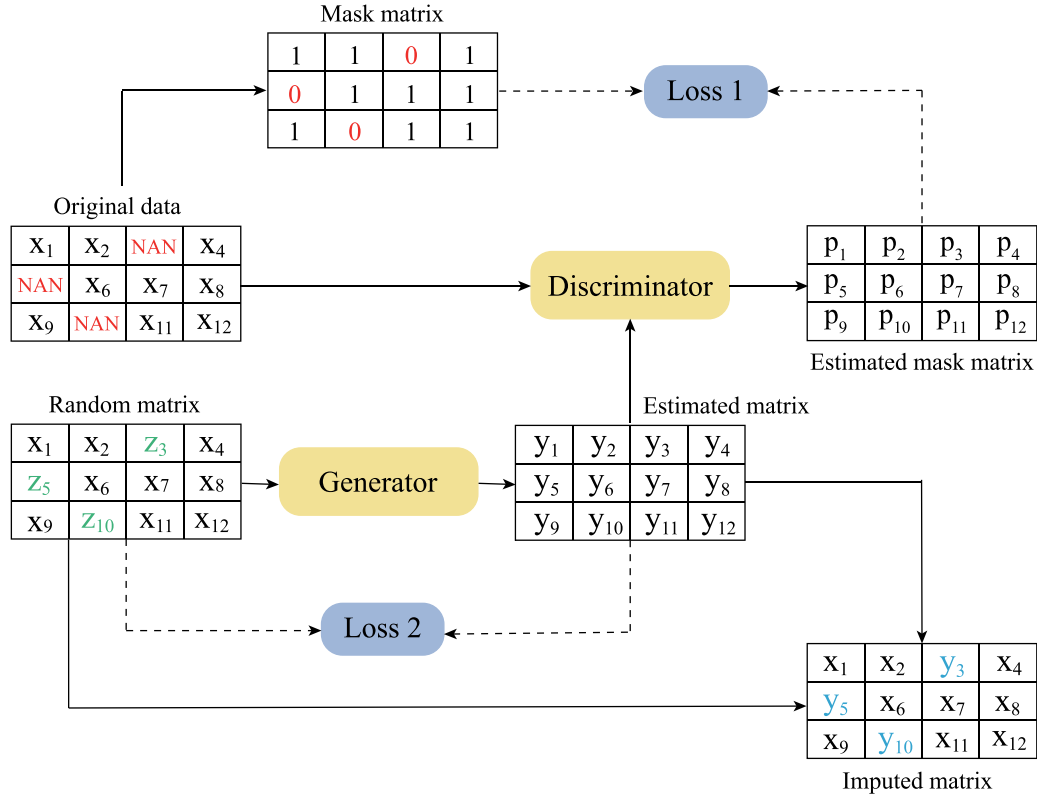


Fig. 3. Specific structure of GAIN.

and D is trained according to $\max_D L_D(m, \hat{m})$. The loss function 2 takes two forms: the first one is given by

$$L_G(m, \hat{m}) = - \sum_i [(1 - m_i) \log(\hat{m}_i)], \quad (5)$$

and the second one is defined by

$$L_M(x, x') = \sum_i m_i (x'_i - x_i)^2. \quad (6)$$

Then, G is trained according to $\min_G [L_G(m, \hat{m}) + \mu L_M(x, x')]$, where μ is a hyper-parameter.

At last, multiple results of each missing attribute value need to be synthesized through weighted average to get the final imputation results. We set the weight as $\alpha_j / \sum_{i=1}^k \alpha_i (1 \leq j \leq k)$. Therefore, the final imputation result is calculated as

$$y = \frac{\alpha_1}{\sum_{i=1}^k \alpha_i} y^{(1)} + \frac{\alpha_2}{\sum_{i=1}^k \alpha_i} y^{(2)} + \dots + \frac{\alpha_k}{\sum_{i=1}^k \alpha_i} y^{(k)}, \quad (7)$$

where $y^{(1)}, y^{(2)}, \dots, y^{(k)}$ are the predicted results of each selected subset.

Through the above description, our MGAIN has the following merits. Firstly, MGAIN can fully consider all the relationships among the missing values, which is helpful to improve the accuracy of imputation results. Meanwhile, it can also effectively impute the multivariate missing data at the same time to improve the imputation efficiency. Then, adding multiple imputation based on GAIN reduces the attribute dimension of dataset, which greatly simplifies the network structure and reduces the demand for data. More importantly, comprehensive consideration of multiple results further improves the accuracy and stability of imputation results.

4. Experiments and analysis

To verify the effectiveness of MGAIN, we carry out three experiments using real data of credit risk assessment, including (1) imputation performance comparison of different methods (see Section 4.2), (2) RMSE performance comparison of methods in different settings (see Section 4.3), (3) RMSE performance comparison between weighted average and arithmetic average in MGAIN (see Section 4.4). In addition, to test the applicability and reliability of MGAIN, we make a comparative experiment using the data of stock price collapse risk (see Section 4.5). All the experimental results demonstrate that the imputation effect of MGAIN achieves state-of-the-art performance in many cases.

The version of the experimental environment is Windows 10 and the system is a 64 bit operating system. The experiments are conducted with Python programming language based on TensorFlow framework. The integrated development environment is PyCharm 2020.3.2 x64, Python version is 3.6.2, and TensorFlow version is 1.15.0.

4.1. Data acquisition and preprocessing

The dataset we adopted for our first three experiments is the private data of credit risk assessment provided by a local bank. Since there are many missing data in the original dataset, we perform some preprocessing at first. After deleting the attributes with 0 value and null value accounting for more than 70%, the final dataset for the experiments includes 10 attributes and 10,744 samples. The 10 attributes mainly include total lending, quarterly average loan balance, annual average loan balance, monthly loan repayments, ratio of household expenditure to income, etc., which are respectively denoted as $A_1, A_2, \dots, A_{10}$. We randomly set the missing data, and the specific missing rate and the number of attributes are introduced in each experiment below.

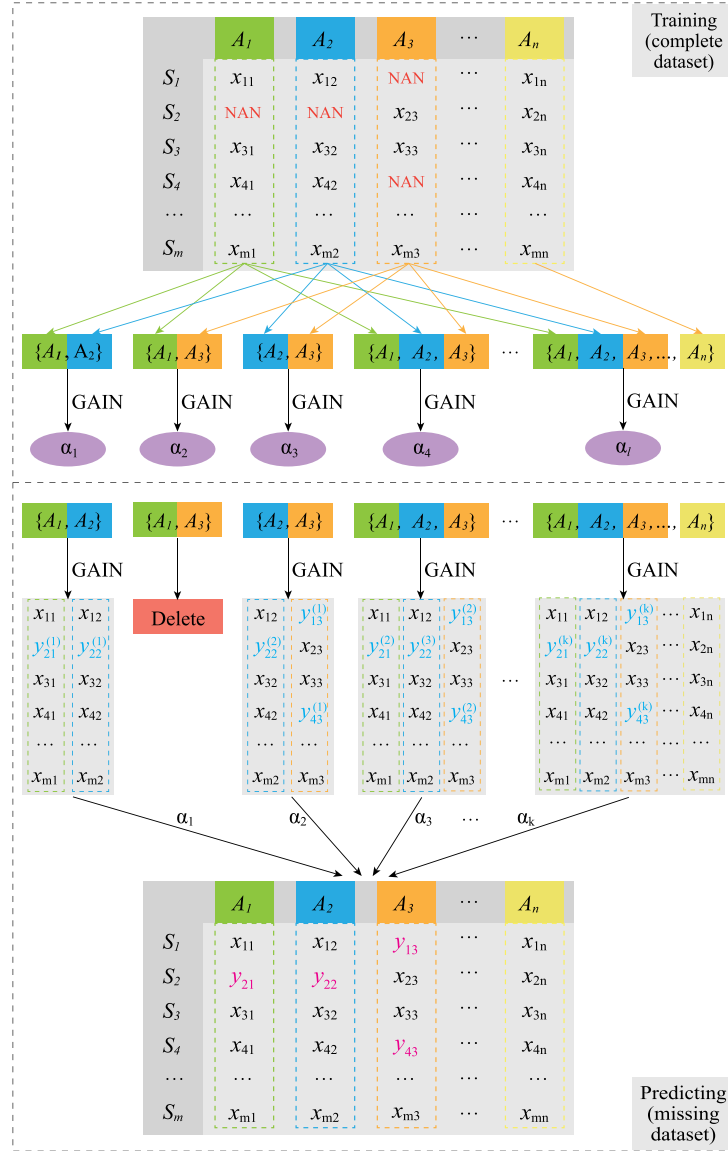


Fig. 4. Specific structure of MGAIN.

The second dataset for our fourth experiment is the financial data related to the risk of stock price collapse, which comes from China Stock Market & Accounting Research Database (CSMAR) and RESSET Database. We select the listed companies in the A-share market of Shanghai and Shenzhen from 2007 to 2017 as the research object. After excluding some inappropriate companies, there are 17,095 samples in total. Referring to relevant research, 10 attributes are selected for the experiment. The attributes mainly include the shareholding ratio of insurance companies, debt-to-asset ratio, return on assets, total assets turnover ratio, Book-to-market ratio and so on. We also randomly set the missing data, and the missing rate is set to 20%.

4.2. Imputation performance comparison of different methods

To demonstrate the imputation performance of MGAIN, we compare the imputation performance of different imputation methods, including MGAIN, GAIN, multiple imputation based on BP neural network (MI-BP), single imputation based on BP neural network (SI-BP), multiple imputation based on linear regression (MI-LR), single imputation based on linear regression (SI-LR) and

mean imputation (MEAN). These comparative methods are popular and widely used in data imputation, and are representative in statistics, machine learning and deep learning respectively. Among them, BP neural network is particularly good at dealing with the nonlinear relationship between data, while linear regression is good at dealing with the linear relationship between data. Each method is tested on the same benchmark, which includes 10,744 samples and 10 attributes, and the missing rate is set to 20%.

For comparing the performance of different imputation methods more intuitively, mean absolute error (MAE) and root mean square error (RMSE) are adopted as evaluation metrics to measure the error of imputation results. Their definitions are given in Table 1, where $\mathbf{Y} = (y_1, y_2, \dots, y_n)$ denotes the real values, $\hat{\mathbf{Y}} = (\hat{y}_1, \hat{y}_2, \dots, \hat{y}_n)$ denotes the imputation values. Smaller values of these metrics indicate better imputation results.

RMSE is also used to measure the prediction effect of subsets, so as to implement subset selection. $1/\text{RMSE}$ is used as an index to measure the prediction accuracy of the selected subsets, which is denoted as $\alpha = (1/e_1, 1/e_2, \dots, 1/e_k)$, where $e = (e_1, e_2, \dots, e_k)$ refers to the RMSE of the selected subsets prediction. Then, the

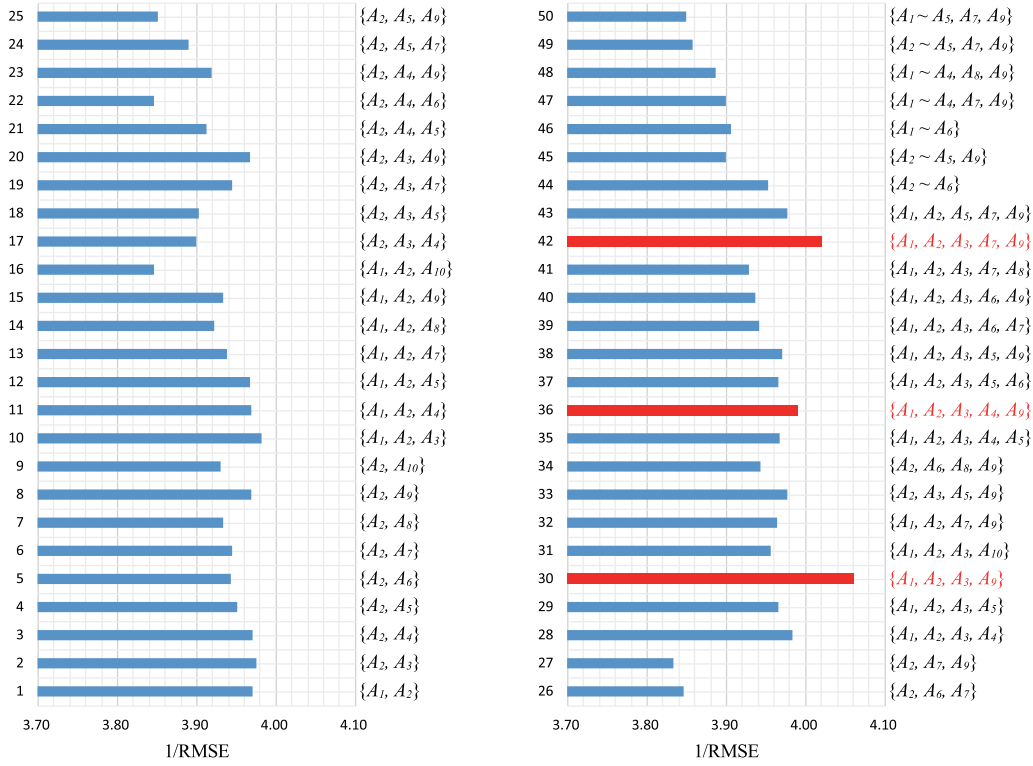
Fig. 5. The top 50 subsets of prediction attribute A_2 .

Table 1

Definitions of two evaluation metrics.

Metrics	Expression
MAE	$\frac{1}{n} \sum_{i=1}^n \hat{y}_i - y_i $
RMSE	$\sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2}$

final imputation result can be expressed as

$$y = \frac{1}{\sum_{i=1}^k \frac{1}{e_i}} y^{(1)} + \frac{1}{\sum_{i=1}^k \frac{1}{e_i}} y^{(2)} + \dots + \frac{1}{\sum_{i=1}^k \frac{1}{e_i}} y^{(k)}, \quad (8)$$

where $y^{(1)}, y^{(2)}, \dots, y^{(k)}$ are the predicted results of each selected subset.

In MGAIN, after training and testing, we select three subsets with the best imputation performance for each attribute to impute the missing values. In order to more intuitively show the process of selecting subsets, we express it through a specific example. Suppose to filter the better performing subsets of prediction attribute A_2 . At first, we can get a total of 511 attribute subsets. Because the sample size of the complete dataset that can be used to train the model in some subsets is small (e.g. less than 2000), we choose to delete these subsets directly. In the remaining subsets, we make a visual comparison of the subsets with the top 50 prediction accuracy. The prediction performance of these 50 subsets on the test set is shown in Fig. 5. We select three subsets with the best prediction effect to impute the missing values in attribute A_2 , namely $\{A_1, A_2, A_3, A_9\}$, $\{A_1, A_2, A_3, A_4, A_9\}$ and $\{A_1, A_2, A_3, A_7, A_9\}$. Similarly, all attributes get three subsets with the best prediction effect, as shown in Table 2. Among them, $\{A_1 \sim A_4\}$ is the abbreviation of $\{A_1, A_2, A_3, A_4\}$.

All selected subsets are directly used for prediction. After the weighted average of multiple results, the final imputation performance of different methods is shown in Table 3. We randomly change the training and testing datasets for 5 times, and

Table 2

Selected subsets of each attribute.

Attribute	Selected subsets
A_1	$\{A_1, A_2, A_3, A_7\}$, $\{A_1, A_2, A_3, A_6, A_8, A_{10}\}$, $\{A_1 \sim A_4, A_6, A_8, A_9\}$
A_2	$\{A_1, A_2, A_3, A_9\}$, $\{A_1 \sim A_4, A_9\}$, $\{A_1, A_2, A_3, A_7, A_9\}$
A_3	$\{A_1, A_2, A_3, A_9\}$, $\{A_1, A_2, A_3, A_5, A_9\}$, $\{A_1, A_2, A_3, A_7, A_9\}$
A_4	$\{A_1, A_4\}$, $\{A_4, A_8\}$, $\{A_4, A_{10}\}$
A_5	$\{A_2, A_3, A_5, A_6, A_7, A_{10}\}$, $\{A_1, A_4 \sim A_7, A_9, A_{10}\}$, $\{A_3 \sim A_8, A_{10}\}$
A_6	$\{A_5 \sim A_8\}$, $\{A_1, A_2, A_4 \sim A_7\}$, $\{A_4 \sim A_9\}$
A_7	$\{A_4 \sim A_8\}$, $\{A_4 \sim A_7, A_9\}$, $\{A_4 \sim A_9\}$
A_8	$\{A_4, A_8\}$, $\{A_1, A_8, A_{10}\}$, $\{A_7, A_8, A_9\}$
A_9	$\{A_3, A_7 \sim A_{10}\}$, $\{A_1, A_2, A_5, A_6, A_7, A_9, A_{10}\}$, $\{A_2, A_3, A_4, A_7 \sim A_{10}\}$
A_{10}	$\{A_3, A_5, A_6, A_8, A_9, A_{10}\}$, $\{A_1, A_3, A_5, A_6, A_7, A_9, A_{10}\}$, $\{A_2 \sim A_{10}\}$

Table 3

Imputation performance comparison of different methods.

	MAE		P value	RMSE		P value
	Average	S.D.		Average	S.D.	
MGAIN	0.0660	0.0008	/	0.1207	0.0007	/
GAIN	0.0715	0.0008	2.6821e-08	0.1269	0.0008	5.9485e-09
MI-BP	0.0729	0.0009	6.5931e-09	0.1293	0.0008	4.4329e-10
SI-BP	0.0740	0.0009	2.0425e-09	0.1315	0.0010	2.1972e-10
MI-LR	0.0733	0.0008	2.8628e-09	0.1299	0.0009	4.0492e-10
SI-LR	0.0748	0.0010	1.7276e-09	0.1332	0.0009	3.5224e-11
MEAN	0.0730	0.0009	5.8837e-09	0.1297	0.0009	4.8228e-10

express the error of the imputation results in the form of average and standard deviation (S.D.). In addition, we conduct statistical significance test (i.e. T-test) for the errors obtained by the six comparison methods and the errors obtained by MGAIN, and the P values of the test are also listed in Table 3. Among them, when P value is less than 0.05, it means that there is a great difference between the tested two.

Table 4
Network structure and calculation of MGAIN and GAIN.

	Number of network layers	Number of neurons per hidden layer	FLOPs	Prediction time (ms)
MGAIN	3	6	648	23.9367
GAIN	5	10	1000	27.9356

Table 3 shows that MAE and RMSE of MGAIN are smallest in all methods, indicating that MGAIN significantly outperforms the other methods. First of all, it can be seen from the Table 3 that the imputation error of GAIN is smaller than single imputation based on BP neural network, single imputation based on linear regression and mean imputation. This can be explained by that GAIN makes full use of the relationships among the missing data, which is helpful to enhance the prediction accuracy. Then, the comparison between MGAIN and GAIN highlights the advantages of subsets selection and multiple imputation. The comprehensive consideration of multiple results makes our final imputation values more convincing and closer to the real values. Finally, all P values in Table 3 are much less than 0.05, which means that the error of MGAIN is significantly different from that of other comparison methods.

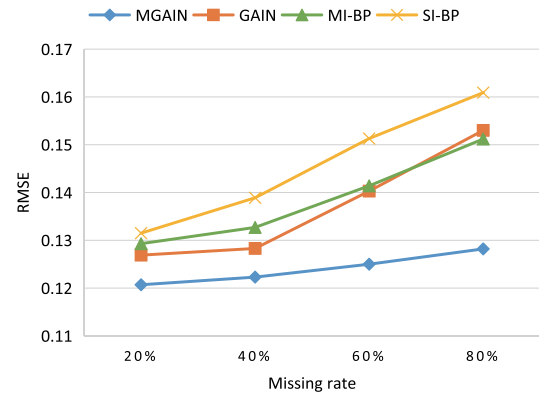
In order to compare the network structure and calculation of MGAIN model and GAIN model, we show the number of network layers, the number of neurons per hidden layer, floating point operations (FLOPs) and prediction time of the two models in Table 4. Among them, FLOPs can measure the complexity of the model, and the smaller its value indicates the less calculation of the model. Note that FLOPs here do not consider the operation of the activation function, and the prediction time is only the time executed on the prediction set. Certainly, the prediction time is directly expressed in the form of the mean value after many experiments.

From Table 4, it can be seen that the number of network layers and the number of neurons per hidden layer in MGAIN are less than that in GAIN, which means that the network structure of generator and discriminator in MGAIN is simpler than that in GAIN under the ability to achieve the highest accuracy. The multiple imputation in MGAIN reduces the number of attributes in the dataset, which leads to a relatively small scale of the trained model. The structural simplification further leads to the calculation amount of the network in MGAIN is much less than that in GAIN, and also improves the operation efficiency.

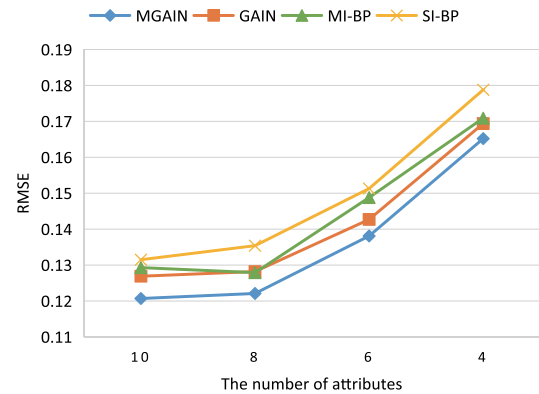
4.3. RMSE performance in different settings

To further evaluate the imputation performance of MGAIN, we conduct several experiments in which we vary the missing rate and the number of attributes. Specifically, in the first experiment, the number of samples and the number of attributes remain unchanged, while the missing rate is set as 20%, 40%, 60% and 80% respectively. In the second experiment, we set the number of attributes as 4, 6, 8 and 10 on the basis of the same sample number and missing rate. Fig. 6 shows the performance (RMSE) of MGAIN with different settings in comparison with the three relevant methods (GAIN, MI-BP and SI-BP). For fair comparison, experiments in different settings were carried out five times, and the mean value of the results was adopted as the final results.

Fig. 6(a) shows the performance of different methods under different missing rate. It can be seen that even though the imputation performance of each method decreases as missing rates increase, MGAIN always outperforms the other three methods at each missing rate. This indicates that MGAIN has high stability, which is less affected by the data missing rate than other methods.



(a) missing rate



(b) the number of attributes

Fig. 6. RMSE performance in different settings: (a) missing rates; (b) the number of attributes.

Fig. 6(b) shows the performance of different methods under different number of attributes. It can be seen that with decrease of the number of attributes, the performance of each method gets worse. However, our MGAIN method still shows relatively good results than the others. The experimental results show that even in the face of dataset with high missing rate or small number of attributes, MGAIN can still result in stable and superior performance.

In addition, we also conduct an experiment to investigate the performance variation of MGAIN under different sample sizes and missing rates. We set four cases of sample size (i.e. 3223, 6446, 9669 and 10744) and four cases of missing rate (i.e. 20%, 40%, 60% and 80%) to compare the RMSE performance of MGAIN and GAIN. The specific operation is the same as the experiment in this section. We can get the imputation errors of the two models in 16 cases, as shown in Fig. 7.

From Fig. 7, it is obvious that the RMSE of both models increases with the decrease of sample size. However, although the sample size is relatively small, the performance of MGAIN is still very stable and better than that of GAIN. Then, when the sample size is the smallest and the missing rate is the highest, the performance of GAIN is worse than MGAIN compared with other cases, which again shows the stable performance of the proposed model. We can analyze that this is because there are few complete datasets available for training when there are serious deficiencies and insufficient samples, resulting in poor prediction effect of GAIN. The subsets selection in MGAIN happens to add

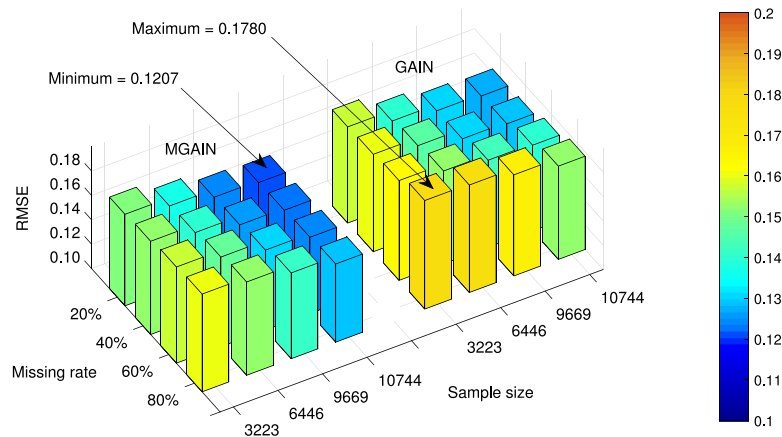


Fig. 7. RMSE performance in different sample sizes and missing rates.

Table 5
RMSE performance comparison of two average methods.

	Subset 1	Subset 2	Subset 3	Weighted average	Arithmetic average
Attribute 1	0.0563	0.0565	0.0569	0.0553	0.0560
Attribute 2	0.2463	0.2487	0.2506	0.2408	0.2451
Attribute 3	0.0166	0.0167	0.0176	0.0152	0.0159

complete datasets, so its prediction performance is not greatly affected.

In order to show the influence of other parameters on the imputation results in MGAIN, we make a sensitivity analysis on the number of network layers, the number of neurons per hidden layer and the number of selected subsets. On the basis of fixed other parameters, we set the number of network layers as 3, 4, 5, 6 and 7 respectively. Similarly, the number of neurons per hidden layer is set as 4, 6, 8, 10 and 12 respectively, and the number of selected subsets is set as 1, 2, 3, 4 and 5 respectively. Fig. 8 shows the imputation errors in different settings of three parameters, where the errors are represented by the mean value.

It is obvious from Fig. 8 that when the number of network layers is 3, the number of neurons per hidden layer is 6, and the number of selected subsets is 3, the error of imputation result is the smallest, i.e., the imputation performance is the best. This is why we choose the set of parameters. Moreover, in most other sets of comparative parameters, the imputation error gap is generally small. In this case, we choose relatively small parameters, which makes the model structure as simple as possible. For example, when the number of selected subsets is 3, 4 and 5, and the imputation performance is roughly the same, we choose the minimum number of subsets.

4.4. RMSE performance comparison of two average methods

In order to explore the effect of the weighted average in MGAIN, we compare it with arithmetic average to calculate the imputation error (RMSE) of three attributes, as shown in Table 5. We also list imputation error of three subsets with the best prediction performance for reference. The missing rate and the number of attributes in this experiment are consistent with the first experiment (see Section 4.2).

Table 5 shows that MGAIN using weighted average achieves the smallest imputation error, indicating that weighted average makes the imputation results much closer to the real values. Weighted average considers the imputation performance of each

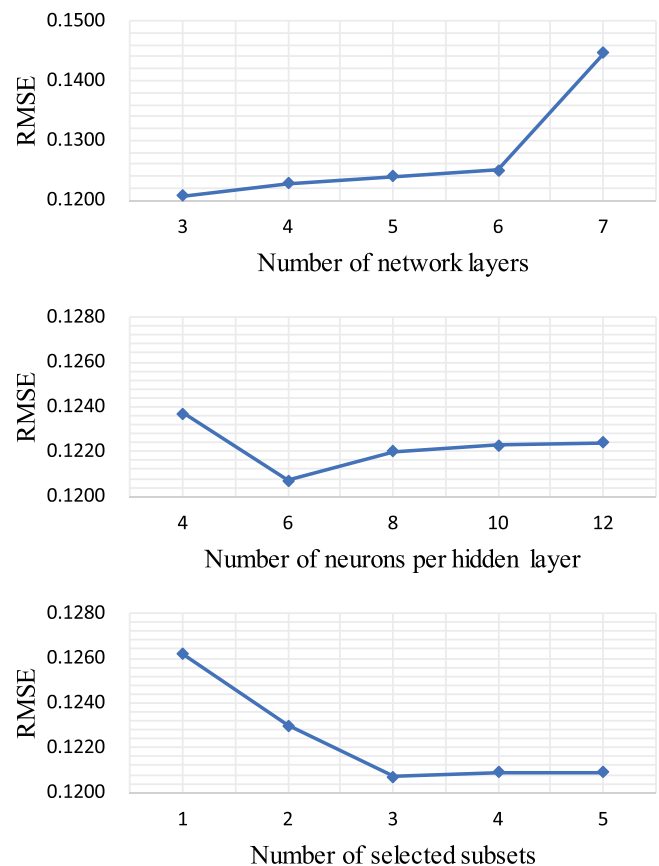


Fig. 8. RMSE performance in different settings of three parameters.

selected subset, and the final imputation result is obtained according to the imputation performance of each subset. Compared with arithmetic average, this method further improves the imputation accuracy. Therefore, in the last step of MGAIN, we choose the weighted average method to synthesize multiple results.

4.5. Imputation performance comparison on new dataset

In order to further demonstrate the applicability and reliability of MGAIN, we utilize a new dataset to compare the imputation performance of different methods. Stock price collapse risk data is similar to credit risk assessment data, both of which have the

Table 6
Imputation performance comparison on new dataset.

	MAE		P value	RMSE		P value
	Average	S.D.		Average	S.D.	
MGAIN	0.2203	0.0007	/	0.3243	0.0008	/
GAIN	0.2256	0.0008	2.0529e−08	0.3319	0.0008	2.0801e−09
MI-BP	0.2258	0.0007	7.9736e−09	0.3322	0.0008	1.5299e−09
SI-BP	0.2281	0.0008	9.6335e−10	0.3357	0.0009	1.2219e−10
MI-LR	0.2279	0.0009	1.8478e−09	0.3352	0.0008	1.1822e−10
SI-LR	0.2310	0.0009	1.2165e−10	0.3408	0.0010	1.1573e−11
MEAN	0.2353	0.0008	5.2556e−12	0.3406	0.0009	7.1745e−13

characteristics of high missing rate and complex data relationship. Thus we use the data of stock price collapse risk to test the imputation performance. Table 6 shows the results achieved by our proposed method and other methods, including GAIN, MI-BP, SI-BP, MI-LR, SI-LR and MEAN. The specific operation process and comparison index display form of the experiment are consistent with those in Section 4.2.

Table 6 shows that MGAIN still maintains excellent imputation performance on the new dataset, which again verifies the applicability and reliability of our proposed method. For datasets that have similar characteristics with credit risk assessment data, MGAIN can also be useful.

5. Conclusion

Dealing with missing data in credit risk assessment is helpful for risk identification, early warning and control, and making better decisions to reduce losses. In this paper, we propose an imputation method, i.e. MGAIN, to accurately predict the missing data in credit risk assessment. Our method first randomly selects multiple subsets from all the attributes, then predicts all the missing values in the subsets by using GAIN, and finally synthesizes multiple prediction results via weighted average to get the final imputation result. Our experiments verify the superiority of MGAIN. On one hand, GAIN makes full use of all the relationships among the missing values to complete the imputation of the multivariate missing model at the same time, which greatly improved the accuracy and efficiency of imputation. On the other hand, subset selection and multiple imputation simplify the network structure of GAIN model and reduce the demand for data. Moreover, the comprehensive consideration of multiple imputation results further improves the accuracy and stability of the final imputation result.

In this article, we only apply MGAIN to missing credit risk assessment data and stock price collapse risk data. Besides, it is also applicable to other datasets with high missing rate and complex missing pattern. Currently, how to effectively screen out the subsets with better imputation performance is still a tough problem. Therefore, our future work will mainly focus on two aspects. First, other suitable datasets will be used to further verify the applicability of the proposed method. Second, a more effective subset filtering method will be explored to improve the imputation accuracy and efficiency.

CRediT authorship contribution statement

Feng Zhao: Conceptualization, Methodology. **Yan Lu:** Conceptualization, Software, Writing – original draft, Methodology, Formal analysis, Investigation, Validation. **Xinning Li:** Validation. **Lina Wang:** Writing – review & editing. **Yingjie Song:** Writing – review & editing. **Deming Fan:** Writing – review & editing. **Caiming Zhang:** Writing – review & editing. **Xiaobo Chen:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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